PRACTICE SHEET

1. If f(x) = A sin and $\int_0^1 f(x)\,dx = \frac{2A}{\pi}$, then what is the value of B?
(a) $2/\pi$ (b) $4/\pi$
(c) 0 (d) 1

2. If m and n are integers, then what is the value of $\int_0^{\pi} \sin mx \sin nx\,dx$, if $m \neq n$?
(a) 0 (b) $\frac{1}{m+n}$
(c) $\frac{1}{m-n}$ (d) mn

3. What is the value of $\int_0^1 (x-1)e^{-x}\,dx$?
(a) 0 (b) e
(c) $\frac{1}{e}$ (d) $\frac{-1}{e}$

4. $\int_{\ln 2}^{x} (e^x - 1)\,dx = \ln\frac{3}{2}$, then what is the value of x?
(a) e^2 (b) $\frac{1}{e}$
(c) In 4 (d) 1

5. If $\int_{-3}^{2} f(x)\,dx = \frac{7}{3}$ and $\int_{-3}^{9} f(x)\,dx = -\frac{5}{6}$, then what is value of $\int_{-3}^{9} f(x)\,dx = -\frac{5}{6}$?
(a) $-\frac{19}{6}$ (b) $\frac{19}{6}$
(c) $\frac{3}{2}$ (d) $-\frac{3}{2}$

6. You are examine these two statements carefully and select the answer.
Assertion (A) : $\int_0^{\pi} \sin^7 x\,dx = 2\int_0^{\pi/2} \sin^7 x\,dx$
Reason (R) : $\sin^7 x$ is an odd function
(a) Both A and R are individually true and R is the correct?
(b) Both A and R are individually true but R is not the correct explanation A
(c) A is true but R is false
(d) A is false but R is true

7. What is the value of integral $\int_0^1 x(1-x)^9\,dx$?
(a) $\frac{1}{110}$ (b) $\frac{1}{111}$
(c) $\frac{1}{112}$ (d) $\frac{1}{119}$

8. What is the value of $\int_{-1}^{1} x\,|x|\,dx$?
(a) 2 (b) 1
(c) $\frac{1}{4}$ (d) 0

9. What is the value of $\int_0^1 \cos^8 x\,dx$?
(a) $\frac{35\pi}{256}$ (b) $\frac{70}{256}$
(c) $\frac{16}{35}$ (d) $\frac{8\pi}{35}$

10. What is $\int_a^b \frac{\log x}{x}\,dx$ equal to?
(a) (1, 2) log (ab). Log (b/a)
(b) log b/log a
(c) log (b/a)
(d) (1/2) log [(a + b)/ab]

11. What is the value of $\int_0^1 x e^{x^2}\,dx$?
(a) $\frac{(e-1)}{2}$ (b) $e^2 - 1$
(c) 2 (e –1) (d) e – 1

12. The value of $\int_{-2}^{2} (ax^3 + bx + c)\,dx$ depends on which of the following?
(a) Value of *x* only
(b) Values of each of *a*, *b* and *c*
(c) Value of *c* only
(d) Value of *b* only

13. What are the values of p which satisfy the question $\int_0^p (3x^2 + 4x - 5)\,dx = p^3 - 2$?
(a) 1 / 2 and 2 (b) –1/ 2 and 2
(c) 1 / 2 and –2 (d) –1 / 2 and – 2

14. What is the value of $\int_0^{\pi/2} \log(\tan x)\,dx$?
(a) 0 (b) 1
(c) –1 (d) $\pi/4$

15. What is $\int_0^1 x(1-x)^n\,dx$ equal to?
(a) $\frac{1}{n(n+1)}$ (b) $\frac{1}{(n+1)(n+2)}$
(c) 1 (d) 0

16. What is the value of the integral $\int_1^2 e^x\left(\frac{1}{x} - \frac{1}{x^2}\right)dx$?
(a) $e\left(\frac{e}{2} - 1\right)$ (b) e(e–1)

(c) $e - \frac{1}{e}$ (d) 0

17. What is the value of the integral $\int_{-1}^{1} |x|\,dx$?

(a) 1 (b) 0
(c) 2 (d) −1

18. What is the value of $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin x \cos x}$?

(a) $2\log\sqrt{3}$ (b) $\log\sqrt{3}$
(c) 2log3 (d) 4log3

19. What is $\int_{-\pi/4}^{\pi/4} \tan^3 x\,dx$ equal to?

(a) $\sqrt{3}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) 0

20. What is $\int_{0}^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x}\,dx$?

(a) π (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) 0

21. What is $\int_{0}^{1} x(1-x)^n\,dx$ equal to?

(a) $\frac{1}{n(n+1)}$ (b) $\frac{1}{(n+1)(n+2)}$
(c) 1 (d) 0

ANSWER KEYS																			
1.	c	**2.**	a	**3.**	d	**4.**	c	**5.**	a	**6.**	b	**7.**	a	**8.**	d	**9.**	a	**10.**	a
11.	a	**12.**	c	**13.**	a	**14.**	a	**15.**	b	**16.**	a	**17.**	a	**18.**	b	**19.**	d	**20.**	c
21.	b																		

Solutions

Sol.1. (c)

Given function $f(x)=A\sin\left(\frac{\pi x}{2}\right)+B$

Differentiating w.r.t.x

$f'(x)=A\cos\left(\frac{\pi x}{2}\right).\frac{\pi}{2}$

$f'\left(\frac{1}{2}\right)=\sqrt{2}=A\left(\cos\frac{\pi}{4}\right)=\frac{\pi}{2}=A.\frac{1}{\sqrt{2}}.\frac{\pi}{2}$

$\Rightarrow A=\frac{\left(\sqrt{2}\times\sqrt{2}\right)\times 2}{\pi}=\frac{4}{\pi}$

Now, $\int_0^1 f(x)dx=\frac{2A}{\pi}$

$\Rightarrow \int_0^1\left\{A\sin\left(\frac{\pi x}{2}\right)+B\right\}dx=\frac{2\times 4}{\pi^2}$

$\Rightarrow \left[-A\cos\frac{\pi x}{2}.\frac{2}{\pi}+Bx\right]_0^1=\frac{8}{\pi^2}$

$\Rightarrow -\frac{4}{\pi}.\frac{2}{\pi}\cos\frac{\pi}{2}+B+\frac{4}{\pi}.\frac{2}{\pi}\cos\theta=\frac{8}{\pi^2}$

$\Rightarrow B+\frac{8}{\pi^2}=\frac{8}{\pi^2}\Rightarrow B=0$

Sol.2. (a)

The given integral

$\int_0^{\pi}\sin mx.\ \mathrm{Sin}\ nx\ dx$

$=\int_0^{\pi}\sin m(\pi-x).\sin n(\pi-x)dx$

$=\int_0^{\pi}\sin mx.\sin nx\ dx$

So, $\int_0^{\pi}\sin mx.\sin nx\ dx$

$=2\int_0^{\pi/2}\sin mx.\sin nx\ dx$

$\int_0^{\pi/2}\frac{1}{2}[\cos(mx-nx)-\cos(mx+nx)]\,dx$

$=\int_0^{\pi/2}[\cos(m-n)x-\cos(m+n)x]dx$

$=\left[\frac{\sin(m-n)x}{m-n}-\frac{\sin(m+n)x}{m+n}\right]_0^{\pi/2}=0$

Sol.3. (d)

Given integral is $I=\int_0^1(x-1)e^{-x}\,dx$

Integrating by parts taking $(x-1)$ as first function

We get, $I=\left[(x-1)\left(-e^{-x}\right)\right]_0^1-\int_0^1 1.\left(-e^{-x}\right)dx$

$=-(1-1)1/e+(-1)e^0+[-e^{-x}]_0^1=-1-\frac{1}{e}+1=-\frac{1}{e}$

Sol.4. (c)

Let $I=\int_{\ln 2}^{x}\left(e^x-1\right)^{-1}dx$

$=\int_{\ln 2}^{x}\frac{1}{e^x-1}dx$

Put $e^x-1=t\Rightarrow e^x=t+1$

$e^x\,dx=dt\Rightarrow dx=\frac{dt}{e^x}$

or, $dx=\frac{dt}{t+1}$

when $x=\ln 2$, $t=e^{\ln 2}-1=2-1=1$

and $I=\int_1^t\frac{1}{t(t+1)}dt$

breaking into partial fractions.

$\frac{1}{t(t+1)}=\frac{t}{t}-\frac{1}{t+1}$

and $I=\int_1^t\left(\frac{1}{t}-\frac{1}{t+1}\right)dt=\left[\log_e t-\log_e(t+1)\right]_1^t$

or $I=\left[\log_e\frac{t}{t+1}\right]_1^t=\log\frac{t}{t+1}-\log_e\frac{1}{2}$

=so, $\frac{2t}{t+1}=\log_e\frac{3}{2}$

$\left[\text{since, }\int_{\ln 2}^{x}\left(e^x-1\right)^{-1}dx=\log_e\frac{3}{2}\right]$

=so, $\frac{2t}{t+1}=\frac{3}{2}$

or, $4t=3t+3\Rightarrow t=3$

so, $e^x-1=3$, $e^x=4\Rightarrow x=\ln 4$

Sol.5. (a)

Value of the integral $\int_2^9 f(x)\,dx$

$\int_{-3}^{9}f(x)\,dx-\int_{-3}^{2}f(x)\,dx$

Given $\int_{-3}^{9}f(x)dx=\frac{-5}{6}$ and $\int_{-3}^{2}f(x)dx=\frac{7}{3}$

Putting these values in equation (i)

$\int_{-3}^{9}f(x)dx=\frac{-5}{6}-\frac{7}{3}=\frac{19}{6}$

Sol.6. (b)

$\int_0^x\sin^7 x\,dx=2\int_0^{x/2}\sin^7 x\,dx$

sin x is an odd function and for an odd function

$\int_0^a f(x)dx=2\int_0^{a/2}f(x)dx$

Hence, $\int_0^x\sin^7 x\,dx=2\int_0^{\pi/2}\sin^7 x\,dx$ is true

Sol.7. (a)

Let the given integral be,

$I=\int_0^1 x(1-x)^9 dx$

Put $1-x=t\Rightarrow dx=-dt$ and $x=1-t$

When $x=0$, $t=1$ and when $x=1$, $t=0$

$\Rightarrow I=\int_0^1(1-t)t^9(-dt)$

$=-\int_0^1\left(-t^{10}+t^9\right)dt=\int_1^0\left(-t^{10}+t^9\right)dt$

$\left[\frac{-t^{11}}{11}+\frac{t^{10}}{10}\right]_0^1=\frac{-1}{11}+\frac{1}{10}=\frac{-10+11}{110}=\frac{1}{110}$

Sol.8. (d)

Let $f(x)=x\,|\,x|$

$f(-x)=-x\,|-x|=-x|x|=-f(x)$

$\Rightarrow x\,|\,x\,|$ is an odd function

Hence $\int_{-1}^{1}x\,|\,x\,|\,dx=0$

Sol.9. (a)

Let, $I=\int_0^{\pi/2}\cos^8 x\,dx$

Given integral can be also be written as:

$I=\int_0^{\pi/2}\sin^{\circ}x$ $\cos^8$ x dx, which is known as Gamma function.

Solution is : $\int_0^{\pi/2}\sin^m x\cos^n x\,dx$

$=\frac{[(m-1)(m-3)....2\text{ or }1][(n-1)(n-3)......2\text{ or }1]}{[(m+n)(m+n-2).....2\text{ or }1]}.\theta$

If m and n both are even then RHS should be multiplied by $\pi/2$here, m = 0. N = 8

$\Rightarrow I=\frac{(8-1)(8-3)(8-5)..(8-7)}{8.(8-2)(8-4)(8-6)}\frac{\pi}{2}$

$=\frac{7.5.3.1}{8.6.4.2}\frac{\pi}{2}=\frac{35\pi}{256}$

Sol.10. (a)

Let the given integer be

$I=\int_a^b\frac{\log x}{x}dx$

Put log x = t and $\frac{dx}{x}$ =dt when x=a, t=log a and if x=b, t=log b

$\therefore I=\int_{\log a}^{\log b}t\,dt=\left[\frac{t^2}{2}\right]_a^b$

$=1/2\,[(\log b)^2-(\log a)^2]$

$=1/2\,[(\log b+\log a)(\log b-\log a)]$

$=1/2\log(ab)\log\left(\frac{b}{a}\right)$

Sol.11. (a)

Let $I=\int_0^1 xe^{x^2}.dx$

Let $x^2=t$

$\Rightarrow 2x\,dx = dt$

$\Rightarrow x dx = \frac{dt}{2}$

when x=0, t=0 then x = 1, t=1

$\Rightarrow I = \frac{1}{2}\int_0^1 e^t dt = \frac{1}{2}\left[e^t\right]_0^1$

$= \frac{1}{2}\left[e^{x^2}\right]_0^1 = \frac{1}{2}\left[e - e^0\right] = \frac{e-1}{2}$

Sol.12. (c)

$\int_{-2}^{2}\left(ax^3 + bx + c\right)dx$

$= \left[\frac{ax^4}{4} + \frac{bx^2}{2} + \frac{cx}{1}\right]_{-2}^{2}$

$= \left[\frac{a(16)}{4} + \frac{b(4)}{2} + 2c\right] - \left[\frac{a(16)}{4} + \frac{b(4)}{2} - 2c\right] = 4c$

So, the value of given integral depends on the value of c only

Sol.13. (a)

Given equation,

$\int_0^p\left(3x^2 + 4x - 5\right)dx = p^3 - 2$

$\Rightarrow \left[\frac{3x^3}{3} + \frac{4x^2}{2} - 5x\right]_0^p = p^3 - 2$

$\Rightarrow p^3+2p^2-5p=p^3-2$

$\Rightarrow 2p^2-5p+2=0$

$\Rightarrow 2p^2-4p-p+2=0$

$\Rightarrow 2p(p-2)-1(p-2)=0$

$\Rightarrow (p-2)(2p-1)=0$

$\Rightarrow p-2=0, 2p-1=0$

Sol.14. (a)

Let $I = \int_0^{\pi/2}\log(\tan x)dx$ …(i)

and $I = \int_0^{\pi/2}\log\left\{\tan\left(\frac{\pi}{2} - x\right)\right\}dx$

[By the property of definite integral which says

$\int_0^a f(x)dx = \int_0^a f(a-x)dx$ …(ii)

$\Rightarrow I = \int_0^{\pi/2}\log(\cot x)dx$

By adding equation (i) and (ii), we get

$2I = \int_2^{\pi/2}\log(\tan x)dx + \int_0^{\pi/2}\log(\cot x)dx$

$2I = \int_2^{\pi/2}\log(\tan x)dx + \int_0^{\pi/2}\log(\cot x)\,dx$

$\Rightarrow 2I = \int_0^{\pi/2}\log(\tan x\cot x)dx$

[log m + log n = log (mn)]

$= \int_0^{\pi/2}\log\left(\tan x.\frac{1}{\tan x}\right)dx = \int_0^{\pi/2}\log 1dx = 0$

$\Rightarrow I = 0$

Sol.15. (b)

Let $I = \int_0^1 x(1-x)^n\,dx$

Put $-x = t \Rightarrow dx = -dt$

When $x = 0$ then $t = 1$

When $x = 1$ then $t = 0$

$\therefore I = -\int_1^0 (1-t)t^n dt = \int_0^1\left(t^n - t^{n+1}\right)dt$

$= \left[\frac{t^{n+1}}{n+1} = \frac{t^{n+2}}{n+2}\right]_0^1$

$= \frac{1}{n+1} = \frac{1}{n+2} = \frac{1}{(n+1)(n+2)}$

Sol.16. (a)

$\int_1^2 e^x\left(\frac{1}{x} - \frac{1}{x^2}\right)dx = \left(\frac{e^x}{x}\right)_1^2 = e\left(\frac{e}{2} - 1\right)$

Sol.17. (a)

$\int_{-1}^{1}|x|\,dx = 2\int_0^1 x dx = 1$

Sol.18. (b)

$\int_{\pi/6}^{\pi/4}\frac{dx}{\sin x\cos x} = 2\int_{\pi/6}^{\pi/4}\frac{dx}{2\sin x\cos x} = 2\int_{\pi/6}^{\pi/4}\frac{dx}{\sin 2x}$

$= 2\int_{\pi/6}^{\pi/4}\operatorname{cosec}2x.dx = 2[\log\tan x]_{\pi/6}^{\pi/4}.\frac{1}{2} = \log\sqrt{3}$

Sol.19. (d)

$\int_{-\pi/4}^{\pi/4}\tan^3 x\,dx = 0$ (odd function)

Sol.20. (c)

$I = \int_0^{\pi/2}\frac{\sin^3 x}{\sin^3 x + \cos^3 x}$

$\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$

$I = \int_0^{\pi/2}\frac{\cos^3 x}{\sin^3 x + \cos^3 x}$

$2I = \int_0^{\pi/2}\frac{\sin^3 x + \cos^3 x}{\sin^3 x + \cos^3 x}dx = \int_0^{\pi/2}1dx = \frac{\pi}{2}$

$I = \frac{\pi}{4}$

Sol.21. (b)

$\int_0^1 x(1-x)^n$

dx $\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$

$= \int_0^1(1-x)(x)^n dx = \int_0^1(x^n - x^{n+1})dx$

$= \int_0^1(x^n - x^{n+1})dx = \left[\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2}\right]_0^1$

$= \left[\frac{1}{n+1} - \frac{1}{n+2}\right] = \frac{1}{(n+1)(n+2)}$

NDA PYQ

1. If f(x) is an even function, then what is $\int_0^{\pi} f(\cos x)dx$ equal to?

(a) 0
(b) $\int_0^{\pi/2} f(\cos x)dx$
(c) $2\int_0^{\pi/2} f(\cos x)dx$
(d) 1

[NDA (I) - 2011]

2. What is $\int_0^{\pi} \frac{dx}{1+2\sin^2 x}$ equal to?

(a) π
(b) $\frac{\pi}{3}$
(c) $\frac{\pi}{\sqrt{3}}$
(d) $\frac{2\pi}{\sqrt{3}}$

[NDA (I) - 2011]

3. If $I_n = \int_0^{\pi/4} \tan^n x dx$, then what is $I_n + I_{n-2}$ equal to?

(a) $\frac{1}{n}$
(b) $\frac{1}{(n-1)}$
(c) $\frac{n}{(n-1)}$
(d) $\frac{1}{(n-2)}$

[NDA (I) - 2011]

4. If $\int_1^2 \{k^2+(4-4k)x+4x^3\}dx \le 12$, then which one of the following is correct?

(a) k=3
(b) $0 \le k < 3$
(c) $k \le 4$
(d) k = 0

[NDA (II)-2011]

5. What is the value of $\int_{-\pi/2}^{\pi/2} |\sin x|\, dx$?

(a) 2
(b) 1
(c) π
(d) 0

[NDA (I) - 2012]

6. What is the value of $\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$ dx?

(a) $\frac{\pi^2}{8}$
(b) $\frac{\pi^2}{32}$
(c) $\frac{\pi}{4}$
(d) $\frac{\pi}{8}$

[NDA (I) - 2012]

7. What is the value of $\int_0^{\pi/2} \sin 2x \ln(\cot x)\, dx$?

(a) 0
(b) $\pi \ln 2$
(c) $-\pi \ln 2$
(d) $\frac{\pi \ln 2}{2}$

[NDA (II) - 2012]

8. What is the value of $\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$ dx?

(a) $\frac{\pi^2}{8}$
(b) $\frac{\pi^2}{32}$
(c) $\frac{\pi}{4}$
(d) $\frac{\pi}{8}$

[NDA (II) - 2012]

9. What is $\int_{-1}^{1} x|x|\, dx$ equal to?

(a) 2
(b) 1
(c) 0
(d) –1

[NDA-2012(2)]

10. What is $\int_0^1 xe^x\, dx$ equal to?

(a) 1
(b) –1
(c) 0
(d) e

[NDA (I) - 2013]

11. What is $\int_0^2 \frac{dx}{x^2+4}$ equal to?

(a) $\frac{\pi}{2}$
(b) $\frac{\pi}{4}$
(c) $\frac{\pi}{8}$
(d) $\frac{\pi}{3}$

[NDA (I) - 2013]

12. What is $\int_{-a}^{a} (x^3+\sin x)\, dx$ equal to?

(a) a
(b) 2a
(c) 0
(d) 1

[NDA (I) - 2013]

13. What is $\int_{-\pi/6}^{\pi/6} \frac{\sin^5 x \cos^3 x}{x^4}\, dx$ equal to?

(a) $\frac{\pi}{2}$
(b) $\frac{\pi}{4}$
(c) $\frac{\pi}{8}$
(d) 0

[NDA (I) - 2013]

14. What is $\int_0^2 e^{\log x}\, dx$ equal to?

(a) 1
(b) 2
(c) 4
(d) None of these

[NDA (II) - 2013]

15. What is $\int_1^2 \log x dx$ equal to?

(a) $\log e^2$
(b) 1
(c) $\log_e\left(\frac{4}{e}\right)$
(d) $\log_e\left(\frac{e}{4}\right)$

[NDA (II) - 2013]

OFFICIUM SUPREMUM

16. What is $\int_{-\pi/2}^{\pi/2} x\sin x\,dx$ equal to?

(a) 0 (b) 2
(c) –2 (d) π

[NDA (I) - 2014]

17. What is $\int_{0}^{\pi/2} \ln(\tan x)dx$ equal to?

(a) In 2 (b) –In 2
(c) 0 (d) None of these

[NDA (I) - 2014]

18. What is $\int_{0}^{1} \frac{e^{\tan^{-1}x}dx}{1+x^2}$ equal to?

(a) $e^{\frac{\pi}{4}}-1$ (b) $e^{\frac{\pi}{4}}+1$
(c) e – 1 (d) e

[NDA (I) - 2014]

Directions (for next two):

$I_1 = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ and $I_2 = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}dx}{\sqrt{\sin x}+\sqrt{\cos x}}$

19. What is $I_1 - I_2$ equal to?

(a) 0 (b) $2I_1$
(c) π (d) None of these

[NDA (I) - 2014]

20. What is I_1 equal to?

(a) $\frac{\pi}{24}$ (b) $\frac{\pi}{18}$
(c) $\frac{\pi}{12}$ (d) $\frac{\pi}{6}$

[NDA (I) - 2014]

Directions (for next two):

Consider the integral $I = \int_{0}^{\pi} \ln(\sin x)dx$

21. What is $\int_{0}^{\pi/2} \ln(\sin x)dx =$

(a) 4I (b) 2I
(c) I (d) $\frac{I}{2}$

[NDA (II) - 2014]

22. What is $\int_{0}^{\pi/2} \ln(\cos x)dx$ equal to?

(a) $\frac{I}{2}$ (b) I
(c) 2I (d) 4I

[NDA (II) - 2014]

23. What is $\int_{0}^{\pi/2} \frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$ equal to?

(a) 2ab (b) 2πab
(c) $\frac{\pi}{2ab}$ (d) $\frac{\pi}{ab}$

[NDA (II) - 2014]

Directions (for next three): Read the following information carefully and answer the questions given below.
For the next three solutions consider:

$I = \int_{0}^{\pi} \frac{xdx}{1+\sin x}$

24. What is I equal to?

(a) –π (b) 0
(c) π (d) 2π

[NDA (II) - 2014]

25. What is $\int_{0}^{\pi} \frac{(\pi-x)dx}{1+\sin x}$ equal to?

(a) π (b) $\frac{\pi}{2}$
(c) 0 (d) 2π

[NDA (II) - 2014]

26. What is $\int_{0}^{\pi} \frac{dx}{1+\sin x}$ equal to?

(a) 1 (b) 2
(c) 4 (d) –2

[NDA (II) - 2014]

Directions (for next four): Consider the integral $I_m = \int_{0}^{\pi} \frac{\sin 2mx}{\sin x}dx$, where m is a positive integer.

27. What is I_1 equal to?

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

[NDA (I) - 2015]

28. What $I_2 + I_3$ equal to?

(a) 4 (b) 2
(c) 1 (d) 0

[NDA (I) - 2015]

29. What is I_m equal to:

(a) 0 (b) 1
(c) m (d) 2m

[NDA (I) - 2015]

30. Consider the following:

I. $I_m + I_{m-1}$ is equal to 0
II. $I_{2m} > I_m$

Which of the above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2015]

31. The value of: $\int_{a}^{b} \frac{x^7+\sin x}{\cos x}dx$ where a + b = 0 is:

(a) 2b –a sin (b–a) (b) a+ 3b cos (b–a)
(c) sin a – (b–a) cos b (d) 0

[NDA (II) - 2015]

32. $\int_{-1}^{1} x\,|x|\,dx$ is equal to:

(a) 0 (b) $\frac{2}{3}$
(c) 2 (d) –2

[NDA (II) - 2015]

33. If 0 < a < b, then $\int_{a}^{b} \frac{|x|}{x}dx$ is equal to?

(a) |b| – |a| (b) |a| – |b|

(c) $\frac{|b|}{|a|}$ (d) 0

[NDA (II) - 2015]

Direction (for next two) :
Consider the integrals

$A = \int_0^{\pi} \frac{\sin x dx}{\sin x + \cos x}$ and $B = \int_0^{\pi} \frac{\sin x dx}{\sin x - \cos x}$

34. Which of the following is correct?
(a) A = 2B (b) B = 2A
(c) A = B (d) A = 3B

[NDA (II) - 2015]

35. What is the value of B ?
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\frac{3\pi}{4}$ (d) π

[NDA (II) - 2015]

36. $\int_0^{2\pi} \sin^5\left(\frac{x}{4}\right) dx$ is equal to:
(a) $\frac{8}{15}$ (b) $\frac{16}{15}$
(c) $\frac{32}{15}$ (d) 0

[NDA (II) - 2015]

37. What is $\int_0^{4\pi} |\cos x| dx$ equal to?
(a) 0 (b) 2
(c) 4 (d) 8

[NDA (I) - 2016]

For the next two (2) items that follow:
Consider the functions

f(x) = xg (x) and g(x) = $\left[\frac{1}{x}\right]$

where [.] is the greatest integer function.

38. What is $\int_{\frac{1}{3}}^{\frac{1}{2}} g(x) dx$ equal to?
(a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{5}{18}$ (d) $\frac{5}{36}$

[NDA (I) - 2016]

39. What is $\int_{\frac{1}{3}}^{1} f(x) dx$ equal to?
(a) $\frac{37}{72}$ (b) $\frac{2}{3}$
(c) $\frac{17}{72}$ (d) $\frac{37}{144}$

[NDA (I) - 2016]

40. What is $\int_{-2}^{2} x\, dx - \int_{-2}^{2} [x] dx$ equal to, where [.] is the greatest integer function?
(a) 0 (b) 1
(c) 2 (d) 4

[NDA (I) - 2016]

41. If $\int_{-2}^{5} f(x) dx = 4$ and $\int_0^5 \{1 + f(x)\} dx = 7$, then what is $\int_{-2}^{0} f(x) dx$ equal to?
(a) –3 (b) 2
(c) 3 (d) 5

[NDA (I) - 2016]

For the next two (2) items that follow:

Given that $a_n = \int_0^{\pi} \frac{\sin^2\{(n+1)x\}}{\sin 2x} dx$

42. Consider the following statements:
1.The sequence $\{a_{2n}\}$ is in AP with common difference zero.
2.The sequence $\{a_{2n+1}\}$ is in AP with common difference zero
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2016(1)]

43. What is $a_{n-1} - a_{n-4}$ equal to?
(a) –1 (b) 0
(c) 1 (d) 2

[NDA-2016(1)]

44. Let f(x) be a function such that:
$f'\left(\frac{1}{x}\right) + x^3 f'(x) = 0$. What is $\int_{-1}^{1} f(x) dx$ equal to?
(a) 2f(1) (b) 0
(c) 2f(–1) (d) 4f(1)

[NDA (II) - 2016]

45. If $\int_0^{\pi/2} \frac{dx}{3\cos x + 5} = k \cot^{-1} 2$, then what is the value of k?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

[NDA (II) - 2016]

46. What is $\int_1^3 \left|1 - x^4\right| dx$ equal to?
(a) $-\frac{232}{5}$ (b) $-\frac{116}{5}$
(c) $\frac{116}{5}$ (d) $\frac{232}{5}$

[NDA (II) - 2016]

47. What is $\int_0^{\pi/2} \frac{d\theta}{1+\cos\theta}$ equal to?
(a) $\frac{1}{2}$ (b) 1
(c) $\sqrt{3}$ (d) None of these

[NDA (I) - 2017]

48. If f(x) and g(x) are continuous functions satisfying f(x) = f(a–x) and g(x) + g (a–x) = 2, then what is $\int_0^a f(x) g(x) dx$ equal to?

(a) $\int_0^a g(x)dx$ (b) $\int_0^a f(x)dx$

(c) $2\int_0^a f(x)dx$ (d) 0

[NDA (I) - 2017]

49. What is $\int_{e^{-1}}^{e^2}\left|\frac{\ln x}{x}\right|dx$ equal to?

(a) $\frac{3}{2}$ (b) $\frac{5}{2}$

(c) 3 (d) 4

[NDA (I) - 2017]

50. What is $\int_0^{2\pi}\sqrt{1+\sin\frac{x}{2}}dx$ equal to?

(a) 8 (b) 4

(c) 2 (d) 0

[NDA (II) - 2017]

51. The value of: $\int_0^{\pi/4}\sqrt{\tan x}dx+\int_0^{\pi/4}\sqrt{\cot x}dx$ is equal to:

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{2\sqrt{2}}$ (d) $\frac{\pi}{\sqrt{2}}$

[NDA (II) - 2017]

52. What is $\int_0^1 x\,(1-x)^9\,dx$ equal to?

(a) 1/110 (b) 1/132

(c) 1/148 (d) 1/240

[NDA (I) - 2018]

53. If $\int_a^b x^3dx = 0$ and $\int_a^b x^2dx = \frac{2}{3}$, then what are the values of a and b respectively?

(a) –1,1 (b) 1,1

(c) 0, 0 (d) 2, –2

[NDA (I) - 2018]

54. What is the value of $\int_{-\pi/4}^{\pi/4}(\sin x-\tan x)dx$?

(a) $-\frac{1}{\sqrt{2}}+\ell n\left(\frac{1}{\sqrt{2}}\right)$ (b) $\frac{1}{\sqrt{2}}$

(c) 0 (d) $\sqrt{2}$

[NDA (I) - 2018]

55. What is $\int_0^{\sqrt{2}}\left[x^2\right]dx$ equal to (where [.] is the greatest integer function)?

(a) $\sqrt{2}-1$ (b) $1-\sqrt{2}$

(c) $2\left(\sqrt{2}-1\right)$ (d) $\sqrt{3}-1$

[NDA (I) - 2018]

56. What is $\int_1^e x\,\ell n\,x\,dx$ equal to:

(a) $\frac{e+1}{4}$ (b) $\frac{e^2+1}{4}$

(c) $\frac{e-1}{4}$ (d) $\frac{e^2-1}{4}$

[NDA (I) - 2018]

57. What is $\int_0^{\pi}e^x\sin x\,dx$ equal to?

(a) $\frac{e^{\pi}+1}{2}$ (b) $\frac{e^{\pi}-1}{2}$

(c) $e^{\pi}+1$ (d) $\frac{e^{\pi}+1}{4}$

[NDA (I) - 2018]

58. What is $\int_{-1}^{1}\left\{\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right)\right\}dx$ equal to?

(a) 0 (b) $-\frac{\pi}{4}$

(c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$

[NDA (II) - 2018]

59. What is $\int_2^8|x-5|\,dx$ equal to?

(a) 2 (b) 3

(c) 4 (d) 9

[NDA (II) - 2018]

60. What is $\int_a^b[x]dx+\int_a^b[-x]dx$ equal to, where [.] is the greatest integer function?

(a) b–a (b) a–b

(c) 0 (d) 2(b–a)

[NDA (II) - 2018]

61. What is $\int_0^{\pi/2}\left|\sin x-\cos x\right|dx$? equal to?

(a) 0 (b) $2(\sqrt{2}-1)$

(c) $2\sqrt{2}$ (d) $2(\sqrt{2}+1)$

[NDA (I) - 2019]

62. What is $\int_0^{\pi/2}e^{\sin x}\cos x dx$? equal to?

(a) e + 1 (b) e – 1

(c) e + 2 (d) e

[NDA (I) - 2019]

Direction (for the next two):
Consider the integrals

$$I_1=\int_0^{\pi}\frac{xdx}{1+\sin x}\text{ and }I_2=\int_0^{\pi}\frac{(\pi-x)dx}{1-\sin(\pi+x)}$$

63. What is the value of I_1?

(a) 2π (b) π

(c) π/2 (d) 2π

[NDA (II) - 2019]

64. What is the value of $I_1 + I_2$?

(a) 2π (b) π

(c) π/2 (d) 2π

[NDA (II) - 2019]

65. What is the value of $\int_0^{\pi/4}\left(\tan^3 x+\tan x\right)dx$?

(a) 1/4 (b) 1/2
(c) 1 (d) 2

[NDA 2020]

66. What is $\int_0^a \frac{f(a-x)}{f(x)+f(a-x)}dx$ equal to?

(a) a (b) 2a
(c) 0 (d) a/2

[NDA (I) - 2021]

67. If $\int_0^a [f(x)+f(-x)]dx = \int_{-a}^a g(x)dx$ then what is g(x) equal to ?

(a) f(x) (b) f(−x) + f(x)
(c) −f(x) (d) none of these

[NDA (I) - 2021]

68. If f(x) = 2^x, then what is $\int_2^{10} \frac{f('x)}{f(x)}dx$ equal to?

(a) 4 In2 (b) In4
(c) In5 (d) 8In2

[NDA (II) - 2021]

69. If $\int_{-2}^0 f(x)dx = k$, then $\int_{-2}^0 |f(x)|dx$ is:

(a) less than k
(b) greater than k
(c) less than or equal to k
(d) greater than or equal to k

[NDA (II) - 2021]

70. If f(x) satisfied f(1) = f(4), then what is $\int_1^4 f'(x)dx$ equal to?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA (II) - 2021]

71. What is $\int_0^{\frac{\pi}{2}} e^{1n(\cos x)}dx$ equal to?

(a) –1 (b) 0
(c) 1 (d) 2

[NDA (II) - 2021]

72. What is $\int_0^{\pi} In\left(\tan\frac{x}{2}\right)dx$ equal to?

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

[NDA (II) - 2021]

73. What is $\int_0^{\frac{\pi}{4}} \frac{dx}{(\sin x+\cos x)^2}$ equal to?

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) $\frac{3}{2}$

[NDA (I) - 2022]

74. What is $\int_{-2}^{-1} \frac{x}{|x|}dx$ equal to?

(a) –2 (b) –1
(c) 1 (d) 2

[NDA (I) - 2022]

75. What is $\int_0^1 \ln\left(\frac{1}{x}-1\right)dx$ equal to?

(a) –1 (b) 0
(c) 1 (d) *l*n2

[NDA 2022 (II)]

76. If $\int_0^{\pi/2}(\sin^4 x+\cos^4 x)dx = k$, then what is the value of $\int_0^{20\pi}(\sin^4 x+\cos^4 x)dx$?

(a) k (b) 10k
(c) 20k (d) 40k

[NDA 2022 (II)]

77. What is $\int_{-\pi/2}^{\pi/2}(e^{\cos x}\sin x+e^{\sin x}\cos x)dx$ equal to?

(a) $\frac{e^2-1}{e}$ (b) $\frac{e^2+1}{e}$
(c) $\frac{1-e^2}{e}$ (d) 0

[NDA 2022 (II)]

Consider the following for the next three (03) items that follow:

Let f(x) = $Pe^x + Qe^{2x} + Re^{3x}$, where, P, q, R are real numbers. Further f(0) = 6, f'(*ln* 3) = 282 and $\int_0^{\ln 2} f(x)dx = 11$.

78. What is the value of Q?

(a) 1 (b) 2
(c) 3 (d) 4

[NDA – 2023 (1)]

79. What is the value of R?

(a)1 (b) 2
(c) 3 (d) 4

[NDA – 2023 (1)]

80. What is f'(0) equal to?

(a) 18 (b) 16
(c) 15 (d) 14

[NDA – 2023 (1)]

Consider the following for the next three (03) items that follows:

Let $I_1 = \int_0^{\pi} \frac{x}{1+\cos^2 x}dx$ and $I_2 = \int_0^{\pi}\frac{1}{1+\sin^2 x}dx$

81. What is the value of $\frac{I_1+I_2}{I_1-I_2}$?

(a) 1 (b) π
(c) π^2 (d) $\frac{\pi+2}{\pi-2}$

[NDA – 2023 (1)]

82. What is the value of $8I_1^2$?

(a) π (b) π^2
(c) π^3 (d) π^4

[NDA – 2023 (1)]

83. What is the value of I_2 ?

(a) $\frac{\pi}{\sqrt{2}}$ (b) $\frac{\pi}{2\sqrt{2}}$
(c) $\frac{3\pi}{2\sqrt{2}}$ (d) $\frac{\pi}{4\sqrt{2}}$

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:

Let $I = \int_a^b \frac{|x|}{x} dx$, a < b

84. What is I equal to when a < 0 < b?
(a) a + b (b) a – b
(c) b – a (d) $\frac{(a+b)}{2}$

[NDA – 2023 (1)]

85. What is I equal to when a < b < 0 ?
(a) a+b (b) a–b
(c) b–a (d) $\frac{(a+b)}{2}$

[NDA – 2023 (1)]

Consider the following for the next (02) items that follow

Let I = $\int_{-2\pi}^{2\pi} \frac{\sin^4 x + \cos^4 x}{1+3^x} dx$

86. What is $\int_0^{\pi} (\sin^4 x + \cos^4 x) dx$ equal to?
(a) $\frac{3\pi}{8}$ (b) $\frac{3\pi}{4}$
(c) $\frac{3\pi}{2}$ (d) 3π

[NDA-2023 (2)]

87. What is I equal to?
(a) 0 (b) $\frac{3\pi}{4}$
(c) $\frac{3\pi}{2}$ (d) 3π

[NDA-2023 (2)]

88. What is $\int_0^{8x} |\sin x| dx$ equal to?
(a) 2 (b) 4
(c) 8 (d) 16

[NDA-2023 (2)]

Consider the following for the next two (02) items that follow:

Let f(x) = |x – 1|, g(x) = [x] and
h(x) = f(x)g(x) where [.] is greatet integer function.

89. What is $\int_{-1}^{0} h(x)dx$ equal to?
(a) $-\frac{3}{2}$ (b) –1
(c) 0 (d) $\frac{1}{2}$

[NDA-2024 (1)]

90. What is $\int_0^2 h(x)dx$ equal to?
(a) $-\frac{3}{2}$ (b) –1
(c) 0 (d) $\frac{1}{2}$

[NDA-2024 (1)]

Consider the following for the next item that follow:

Let $3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1$

91. What is $8\int_1^2 f(x)dx$ equal to ?
(a) $\ln(8\sqrt{e})$ (b) $\ln(4\sqrt{e})$
(c) ln 2 (d) ln 2 – 1

[NDA-2024 (1)]

92. What is $\int_{-1}^{1} (3\sin x - \sin 3x)\cos^2 x dx$ equal to?
(a) $-\frac{1}{4}$ (b) 0
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

[NDA-2024 (1)]

93. Let $p = \int_a^b f(x)dx$ and $q = \int_a^b |f(x)| dx$.
If f(x) = e^{-x}, then which one of the following is correct?
(a) p = 2q (b) p = –q
(c) 4p = q (d) p = q

[NDA-2024 (1)]

94. What is $\int_0^{\frac{\pi}{2}} \frac{a+\sin x}{2a+\sin x+\cos x} dx$ equal to?
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) 1 (d) 0

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

Let $\varphi(a) = \int_a^{a+100\pi} |\sin x| dx$

95. What is φ(a) equal to?
(a) 0 (b) a
(c) 100a (d) 200

[NDA-2024 (1)]

96. What is φ′(a) equal to?
(a) 0 (b) π
(c) 100 (d) 200

[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

Let f(x) = $|x^2 - x - 2|$

97. What is $\int_0^2 f(x)dx$ equal to ?
(a) 0 (b) 1
(c) 5/3 (d) 10/3

[NDA-2024 (2)]

98. What is $\int_1^3 f(x)dx$ equal to ?
(a) 2 (b) 3
(c) 4 (d) 5

[NDA-2024 (2)]

Direction: Consider the following for the **one items** given below

Let $f(t)=\ln\left(t+\sqrt{1+t^2}\right)$ and $g(t)=\tan\left(f(t)\right)$

99. What is $\int_{-\pi}^{\pi} g(t)dt$ equal to ?

(a) –1 (b) 0

(c) 1/2 (d) 1

[NDA-2024 (2)]

Direction: Consider the following for the **two items** given below

Let $I=\int_0^{\pi/2}\frac{f(x)}{g(x)}dx$, where *f(x) = sinx* and

g(x) = sinx + cosx + 1

100. What is $\int_0^{\pi/2}\frac{dx}{g(x)}$ equal to?

(a) $\frac{\ln 2}{2}$ (b) $\frac{\ln 2}{4}$

(c) ln2 (d) 2ln2

[NDA-2024 (2)]

101. What is *I* equal to ?

(a) $\frac{\pi}{4}+\ln 2$ (b) $\frac{\pi}{4}-\ln 2$

(c) $\frac{\pi}{4}-\frac{\ln 2}{2}$ (d) $\frac{\pi}{4}+\frac{\ln 2}{2}$

[NDA-2024 (2)]

Consider the following for the two (02) items that follow:

Let $f(x)=\left[x^2\right]$, where [.] is the greatest integer function.

102. What is $\int_{\sqrt{2}}^{\sqrt{3}} f(x)dx$ equal to?

(a) $\sqrt{3}-\sqrt{2}$ (b) $2\left(\sqrt{3}-\sqrt{2}\right)$

(c) $3-\sqrt{2}$ (d) 1

[NDA-2025 (1)]

103. What is $\int_{\sqrt{2}}^{2} f(x)dx$ equal to?

(a) $6-\sqrt{3}-2\sqrt{2}$ (b) $6-\sqrt{3}-\sqrt{2}$

(c) $6-\sqrt{3}+2\sqrt{2}$ (d) $6+\sqrt{3}-2\sqrt{2}$

[NDA-2025 (1)]

104. What is $\int_n^{n+1}\left(x-[x]\right)dx$, where [.] is the greatest integer function and n is natural number?

(a) $\frac{4n+1}{2}$ (b) $\frac{2n+1}{2}$

(c) ½ (d) 1

[NDA-2025 (2)]

105. If $I_1=\int_e^{e^2}\frac{dx}{\ln x}$ and $I_2=\int_1^2\frac{e^x}{x}dx$, then which one of the following is correct?

(a) $I_1 - I_2 = 0$ (b) $I_1 + I_2 = 0$

(c) $I_1 - 2I_2 = 0$ (d) $2I_1 - I_2 = 0$

[NDA-2025 (2)]

ANSWER KEY																			
1.	c	**2.**	c	**3.**	b	**4.**	a	**5.**	a	**6.**	b	**7.**	a	**8.**	b	**9.**	c	**10.**	a
11.	c	**12.**	c	**13.**	d	**14.**	b	**15.**	c	**16.**	b	**17.**	c	**18.**	a	**19.**	a	**20.**	c
21.	d	**22**	a	**23.**	c	**24.**	c	**25.**	a	**26.**	b	**27.**	a	**28.**	d	**29.**	a	**30.**	a
31.	d	**32.**	a	**33.**	a	**34.**	c	**35.**	b	**36.**	c	**37.**	d	**38.**	b	**39.**	a	**40.**	c
41.	b	**42.**	c	**43.**	b	**44.**	c	**45.**	b	**46.**	d	**47.**	b	**48.**	b	**49.**	b	**50.**	a
51.	d	**52.**	a	**53.**	a	**54.**	c	**55.**	a	**56.**	b	**57.**	b	**58.**	d	**59.**	d	**60.**	b
61.	b	**62.**	b	**63.**	b	**64.**	d	**65.**	b	**66.**	d	**67.**	a	**68.**	d	**69.**	d	**70.**	d
71.	c	**72.**	a	**73.**	b	**74.**	b	**75.**	b	**76.**	d	**77.**	a	**78.**	b	**79.**	c	**80.**	d
81.	d	**82.**	d	**83.**	a	**84.**	a	**85.**	b	**86.**	b	**87.**	c	**88.**	d	**89.**	a	**90.**	d
91.	a	**92.**	b	**93.**	d	**94.**	a	**95.**	d	**96.**	a	**97.**	d	**98.**	b	**99.**	b	**100.**	c
101.	c	**102.**	b	**103.**	a	**104.**	c	**105.**	a										

OFFICIUM SUPREMUM

Solutions

Sol.1. (c)

$$\int_0^{\pi} f(\cos x)dx = \int_0^{\pi} f(\cos(\pi - x))dx = \int_0^{\pi} f(-\cos x)dx$$

$$= \int_0^{\pi} f(-\cos x)dx = \int_0^{\pi} f(\cos x)dx$$

so $\int_0^{\pi} f(\cos x)dx = 2\int_0^{\pi/2} f(\cos x)dx$

Sol.2. (c)

$$\int_0^{\pi} \frac{dx}{1+2\sin^2 x} = 2\int_0^{\pi/2} \frac{dx}{1+2\sin^2 x}$$

$$= \int_0^{\pi/2} \frac{\sec^2 x dx}{\sec^2 x + 2\tan^2 x} = \int_0^{\pi/2} \frac{\sec^2 x dx}{1+3\tan^2 x}$$

put tanx = t

$$= \int_0^{\infty} \frac{dt}{1+3t^2} = \frac{1}{3}\int_0^{\infty} \frac{dt}{\left(\frac{1}{\sqrt{3}}\right)^2 + t^2} = \frac{\pi}{\sqrt{3}}$$

Sol.3. (b)

$$I_n = \int_0^{\pi/4} \tan^n x dx$$

$$I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x dx + \int_0^{\pi/4} \tan^2 x \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x(1+\tan^2 x)dx = \int_0^{\pi/4} \tan^{n-2} x(\sec^2 x)dx$$

put tanx = t

$$= \int_0^1 t^{n-2} dt = \left[\frac{t^{n-2+1}}{n-2+1}\right]_0^1 = \frac{1}{n-1}$$

Sol.4. (a)

$$\int_1^2 \{k^2 + (4-4k)x + 4x^3\}dx \le 12,$$

$$\left(k^2 x + (4-4k)\frac{x^2}{2} + 4\frac{x^4}{4}\right)_1^2 \le 12$$

$$\left(k^2 x + (2-2k)x^2 + x^4\right)_1^2 \le 12$$

$$\left(k^2 + 3(2-2k) + 15\right) \le 12$$

$$\left(k^2 - 6k + 21\right) \le 12$$

$$k^2 - 6k + 9 \le 0$$

$$(k-3)^2 \le 0$$

k = 3

Sol.5. (a)

$$\int_{-\pi/2}^{\pi/2} |\sin x|\, dx = 2 \quad \text{(by graph)}$$

Sol.6. (a)

$$\int_0^1 xe^x dx = [xe^x - e^x]_0^1 = 1$$

Sol.7. (c)

$$\int_0^2 \frac{dx}{x^2+4} = \frac{1}{2}\left[\tan^{-1}\frac{x}{2}\right]_0^2 = \frac{\pi}{8}$$

Sol.8. (c)

$$\int_{-a}^{a} \left(x^3 + \sin x\right) dx = 0 \quad \text{(odd function)}$$

Sol.9. (d)

$$\int_{-\pi/6}^{\pi/6} \frac{\sin^5 x \cos^3 x}{x^4} dx = 0 \qquad \text{(odd function)}$$

Sol.10. (a)

$$I = \int_0^{\pi/2} \sin 2x . \text{In}(\cot x) dx$$

$$\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$I = \int_0^{\pi/2} \sin 2x . \text{In}(\tan x) dx$$

$$2I = \int_0^{\pi/2} \sin 2x . \text{In}(\tan x \cot x) dx = 0$$

Sol.11. (b)

$$\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$$

put $\tan^{-1}x = t$

$$\int_0^{\pi/4} t dt = \left[\frac{t^2}{2}\right]_0^{\pi/4} = \frac{\pi^2}{32}$$

Sol.12. (c)

x^3 and sinx both are odd functions so integration of both functions is zero.

Sol.13. (d)

$\sin^5 x$ is odd functions so integration of function is zero.

Sol.14. (b)

$$\int_0^2 e^{\log x} dx = \int_0^2 x dx = \left[\frac{x^2}{2}\right]_0^2 = 2$$

Sol.15. (c)

$$\int_1^2 \log x dx = [x \log x - x]_1^2 = 2\log 2 - 1$$

$$= \log 4 - \log e = \log\left(\frac{4}{e}\right)$$

Sol.16. (b)

$$\int_{-\pi/2}^{\pi/2} x \sin x\, dx = 2\int_0^{\pi/2} x \sin x\, dx$$

(even function)

$$= 2[-x\cos x + \sin x]_0^{\pi/2} = 2$$

Sol.17. (c)

$$I = \int_0^{\pi/2} \text{In}(\tan x) dx \quad \left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$I = \int_0^{\pi/2} \text{In}(\cot x) dx$$

$$2I = \int_0^{\pi/2} \text{In}(\tan x) dx + \int_0^{\pi/2} \text{In}(\cot x) dx$$

$$2I = \int_0^{\pi/2} \text{In}(\tan x \cot) dx = 0$$

Sol.18. (a)

$$\int_0^1 \frac{e^{\tan^{-1} x} dx}{1+x^2}$$

put $\tan^{-1} x = t$

$$\int_0^{\pi/4} e^t dt = e^{\pi/4} - 1$$

Sol.19. (a)

$$I_1 = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x} dx}{\sqrt{\sin x} + \sqrt{\cos x}} \text{ ....(i)}$$

$$\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$= \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x} dx}{\sqrt{\sin x} + \sqrt{\cos x}} = I_2 \text{ .....(ii)}$$

$I_1 - I_2 = 0$

Sol.20. (c)

(i) + (ii)

$$I + I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x} + \sqrt{\cos x}\ dx}{\sqrt{\sin x} + \sqrt{\cos x}} = \int_{\pi/6}^{\pi/3} 1.dx$$

$$2I = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$I = \frac{\pi}{12}$$

Sol.21. (d)

$$I = \int_0^{\pi} \text{In}(\sin x) dx$$

$$\left[\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx \ \text{ when } f(2a-x) = f(x)\right]$$

$$I = 2\int_0^{\pi/2} \text{In}(\sin x) dx$$

$$\int_0^{\pi/2} \text{In}(\sin x) dx = \frac{I}{2}$$

Sol.22. (a)

$$\int_0^{\pi/2} \text{In}(\sin x) dx = \frac{I}{2}$$

$$\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$\int_0^{\pi/2} \text{In}(\cos x) dx = \frac{I}{2}$$

Sol.23. (c)

$$\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x + b^2 \sin^2 x}$$

$$= \int_0^{\pi/2} \frac{\sec^2 x dx}{a^2 + b^2\tan^2 x} = \frac{1}{b^2}\int_0^{\pi/2} \frac{\sec^2 x dx}{\left(\frac{a}{b}\right)^2 + \tan^2 x}$$

put tanx = t

$$= \frac{1}{b^2}\int_0^{\infty}\frac{dt}{\left(\frac{a}{b}\right)^2+t^2} = \frac{1}{ab}\left[\tan^{-1}\left(\frac{x}{a/b}\right)\right]_0^{\infty} = \frac{\pi}{2ab}$$

Sol.24. (c)

$$I_1 = \int_0^{\pi}\frac{xdx}{1+\sin x}.............(i)$$

$$\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$I_1 = \int_0^{\pi}\frac{(\pi-x)dx}{1+\sin(\pi-x)} = \int_0^{\pi}\frac{(\pi-x)dx}{1+\sin x}...........(ii)$$

(i) + (ii)

$$2I_1 = \int_0^{\pi}\frac{\pi dx}{1+\sin x} = \pi\int_0^{\pi}\frac{dx}{1+\sin x} \quad ...(iii)$$

$$\left[\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx \text{ when } f(2a-x) = f(x)\right]$$

$$2I_1 = 2\pi\int_0^{\pi/2}\frac{dx}{1+\sin x}$$

$$2I_1 = 2\pi\int_0^{\pi/2}\frac{dx}{1+\cos\left(\frac{\pi}{2}-x\right)} = 2\pi\int_0^{\pi/2}\frac{dx}{2\cos^2\left(\frac{\pi}{4}-\frac{x}{2}\right)}$$

$$I_1 = \pi\int_0^{\pi/2}\frac{\sec^2\left(\frac{\pi}{4}-\frac{x}{2}\right)dx}{2} = \frac{\pi}{2}\int_0^{\pi/2}\sec^2\left(\frac{\pi}{4}-\frac{x}{2}\right)dx$$

$$I_1 = \frac{\pi}{2}\left[\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right]_0^{\pi/2}\times(-2)$$

$$I_1 = \frac{\pi}{2}[0-1]\times(-2) = \pi$$

Sol.25. (a)

by equation(ii)

$$I_1 = \int_0^{\pi}\frac{(\pi-x)dx}{1+\sin x} = \int_0^{\pi}\frac{xdx}{1+\sin x} = \pi$$

Sol.26. (b)

by equation (iii)

$$2I_1 = \pi\int_0^{\pi}\frac{dx}{1+\sin x}$$

$$\int_0^{\pi}\frac{dx}{1+\sin x} = 2$$

Sol.27. (a)

$$I_1 = \int_0^{\pi}\frac{\sin 2x}{\sin x}dx = \int_0^{\pi}\frac{2\sin x\cos x}{\sin x}dx$$

$$I_1 = \int_0^{\pi}2\cos x dx = 2[\sin x]_0^{\pi} = 0$$

Sol.28. (d)

$$I_m = \int_0^{\pi}\frac{\sin 2mx}{\sin x}dx = \int_0^{\pi}\frac{\sin 2m(\pi-x)}{\sin x}dx \quad .....(i)$$

$$I_m = -\int_0^{\pi}\frac{\sin 2mx}{\sin x}dx \quad ....(ii)$$

(i) + (ii)

$2I_m = 0 \Rightarrow I_m = 0$

so $I_1 = I_2 = I_3 = 0$

Sol.29. (a)

$$I_m = \int_0^{\pi}\frac{\sin 2mx}{\sin x}dx = \int_0^{\pi}\frac{\sin 2m(\pi-x)}{\sin x}dx \quad .....(i)$$

$$I_m = -\int_0^{\pi}\frac{\sin 2mx}{\sin x}dx \quad ....(ii)$$

(i) + (ii)

$2I_m = 0 \Rightarrow I_m = 0$

Sol.30. (a)

$I_m = 0$ so $I_{2m} > I_m$ is wrong

Sol.31. (d)

$$\int_a^b\frac{x^7+\sin x}{\cos x}dx = \int_{-b}^{b}\frac{x^7+\sin x}{\cos x}dx = 0$$

$$\left[\int_{-a}^{a} f(x)dx = 0 \quad \text{if} \quad f(-x) = -f(x)\right]$$

Sol.32. (a)

$$\int_{-1}^{1}x|x|dx = 0$$

$$\left[\int_{-a}^{a} f(x)dx = 0 \quad \text{if} \quad f(-x) = -f(x)\right]$$

Sol.33. (a)

$$\int_a^b\frac{|x|}{x}dx = \int_a^b 1.dx = b-a$$

Sol.34. (c)

$$A = \int_0^{\pi}\frac{\sin xdx}{\sin x+\cos x}$$

$$\left[\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$A = \int_0^{\pi}\frac{\sin(\pi-x)dx}{\sin(\pi-x)+\cos(\pi-x)} = \int_0^{\pi}\frac{\sin xdx}{\sin x-\cos x} = B$$

Sol.35. (b)

$$\int_0^{\pi}\frac{\sin xdx}{\sin x-\cos x} = \frac{1}{2}\int_0^{\pi}\frac{2\sin xdx}{\sin x-\cos x}$$

$$= \frac{1}{2}\int_0^{\pi}\frac{\sin x+\cos x+\sin x-\cos xdx}{\sin x-\cos x}$$

$$= \frac{1}{2}\int_0^{\pi}\frac{(\sin x+\cos x)dx}{\sin x-\cos x}+\frac{1}{2}\int_0^{\pi}\frac{(\sin x-\cos x)dx}{\sin x-\cos x}$$

$$= \frac{1}{2}\int_0^{\pi}\frac{(\sin x+\cos x)dx}{\sin x-\cos x}+\frac{1}{2}\int_0^{\pi}1.dx$$

$= 0 + \pi/2$

Sol.36. (c)

$$\int_0^{2\pi}\sin^5\left(\frac{x}{4}\right)dx = \int_0^{2\pi}\sin^4\left(\frac{x}{4}\right)\sin\left(\frac{x}{4}\right)dx$$

$$= \int_0^{2\pi}\left(1-\cos^2\frac{x}{4}\right)\left(1-\cos^2\frac{x}{4}\right)\sin\left(\frac{x}{4}\right)dx$$

put $\cos\left(\frac{x}{4}\right) = t$

$$= -4\int_1^0(1-t^2)(1-t^2)dt = -4\int_1^0(t^4-2t^2+1)dt$$

$= 32/15$

Sol.37. (d)

$$\int_0^{4\pi}|\cos x|dx = 8 \text{ (by graph)}$$

Sol.38. (b)

$$\int_{\frac{1}{3}}^{\frac{1}{2}}\left[\frac{1}{x}\right]dx = \int_{\frac{1}{3}}^{\frac{1}{2}}2dx = 2\left[\frac{1}{2}-\frac{1}{3}\right] = \frac{1}{3}$$

Sol.39. (a)

$$\int_{\frac{1}{3}}^{1}f(x)dx = \int_{1/3}^{1/2}xg(x)dx+\int_{\frac{1}{3}}^{1}xg(x)dx$$

$$= \int_{1/3}^{1/2}x(2)dx+\int_{\frac{1}{3}}^{1}x.1dx$$

$$= \left[x^2\right]_{1/3}^{1/2}+\left[\frac{x^2}{2}\right]_{1/2}^{1} = \frac{5}{36}+\frac{3}{8} = \frac{37}{72}$$

Sol.40. (c)

$$\int_{-2}^{2}x\,dx-\int_{-2}^{2}[x]dx$$

$$= 0-\int_{-2}^{-1}[x]dx-\int_{-1}^{0}[x]dx-\int_0^1[x]dx-\int_1^2[x]dx$$

$$= -\int_{-2}^{-1}(-2)dx-\int_{-1}^{0}(-1)dx-\int_0^1(0)dx-\int_1^2(1)dx$$

$= 2 + 1 + 1 = 4$

Sol.41. (b)

$$\int_{-2}^{0}f(x)dx+\int_0^5 f(x)dx = 4 \quad ....(i)$$

given $\int_0^5 1dx+\int_0^5\{f(x)\}dx = 7$,

$$= 5+\int_0^5\{f(x)\}dx = 7$$

$$= \int_0^5\{f(x)\}dx = 2$$

now by (i)

$$\int_{-2}^{0}f(x)dx+2 = 4 \Rightarrow \int_{-2}^{0}f(x)dx = 2$$

$\int_0^5\{1+f(x)\}dx$ then what is $\int_{-2}^{0}f(x)dx$ equal to?

Sol.42. (c)

$$a_{2n} = \int_0^{\pi}\frac{\sin^2\{(2n+1)x\}}{\sin 2x}dx \quad ....(i)$$

$$a_{2n} = \int_0^{\pi}\frac{\sin^2\{(2n+1)(\pi-x)\}}{\sin 2(\pi-x)}dx = \int_0^{\pi}\frac{\sin^2\{(2n+1)x\}}{-\sin 2x}dx \quad ......(ii)$$

add both equations

$a_{2n} = 0$

all terms are zero so sequence is in AP with difference 0

same as above

$a_{2n+1} = 0$

all terms are zero so sequence is in AP with difference 0

Sol.43. (b)

$a_{n-1} = a_{n-4}$

Sol.44. (c)

$$f'\left(\frac{1}{x}\right) = -x^3f'(x)$$

divide by x^2

$$-\frac{1}{x^2}f'\left(\frac{1}{x}\right) = xf'(x)$$

integrate both sides

$$-\int_{-1}^{1}\frac{1}{x^2}f'\left(\frac{1}{x}\right)dx=\int_{-1}^{1}xf'(x)dx$$

let 1/x = t in left hand side and integrate right hand side by parts

$$-\int_{-1}^{1}f'(t)dt=[xf(x)]_{-1}^{1}-\int_{-1}^{1}f(x)dx$$

$$[f(t)]_{-1}^{1}=[xf(x)]_{-1}^{1}-\int_{-1}^{1}f(x)dx$$

$$f(1)-f(-1)=f(1)+f(-1)-\int_{-1}^{1}f(x)dx$$

$$\int_{-1}^{1}f(x)dx=2f(-1)$$

Sol.45. (b)

$$\int_0^{\pi/2}\frac{dx}{3\cos x+5}=\int_0^{\pi/2}\frac{dx}{3\left[\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right]+5}$$

$$=\int_0^{\pi/2}\frac{\left[1+\tan^2\frac{x}{2}\right]dx}{3\left[1-\tan^2\frac{x}{2}\right]+5\left[1+\tan^2\frac{x}{2}\right]}$$

$$=\frac{1}{2}\int_0^{\pi/2}\frac{\left[\sec^2\frac{x}{2}\right]dx}{\tan^2\frac{x}{2}+2^2}$$

put $\tan\frac{x}{2}=t$

$$=\int_0^1\frac{dx}{t^2+2^2}=\frac{1}{2}\left[\tan^{-1}\left(\frac{t}{2}\right)\right]_0^1=\frac{1}{2}\tan^{-1}\left(\frac{1}{2}\right)$$

$$=\frac{1}{2}\cot^{-1}2$$

k = ½

Sol.46. (d)

$$\int_1^3|1-x^4|dx=\int_1^3-(1-x^4)dx$$

$$\left[\frac{x^5}{5}-x\right]_1^3=\left[\frac{3^5}{5}-3\right]-\left[\frac{1}{5}-1\right]=\frac{232}{5}$$

Sol.47. (b)

$$\int_0^{\pi/2}\frac{d\theta}{1+\cos\theta}=\int_0^{\pi/2}\frac{d\theta}{2\cos^2\left(\frac{\theta}{2}\right)}=\frac{1}{2}\int_0^{\pi/2}\sec^2\left(\frac{\theta}{2}\right)d\theta$$

$$\frac{1}{2}\int_0^{\pi/2}2\tan\left(\frac{\theta}{2}\right)d\theta=1$$

Sol.48. (b)

$$I=\int_0^a f(x)g(x)dx=\int_0^a f(a-x)g(a-x)dx$$

$$I=\int_0^a f(x)[2-g(x)]dx$$

$$I=\int_0^a[2f(x)-f(x)g(x)]dx$$

$$I=\int_0^a 2f(x)dx-\int_0^a f(x)g(x)dx$$

$$I=\int_0^a 2f(x)dx-I$$

$$2I=\int_0^a 2f(x)dx\Rightarrow I=\int_0^a f(x)dx$$

Sol.49. **(b)**

$$\int_{e^{-1}}^{e^2}\left|\frac{\ln x}{x}\right|$$

$$dx=\int_{e^{-1}}^{1}-\left(\frac{\ln x}{x}\right)dx+\int_1^{e}\left(\frac{\ln x}{x}\right)dx$$

$$=-\frac{1}{2}\left[(\log x)^2\right]_{1/e}^{1}+\frac{1}{2}\left[(\log x)^2\right]_1^{e^2}$$

$$=\frac{1}{2}+\frac{4}{2}=\frac{5}{2}$$

Sol.50. **(a)**

$$\int_0^{2x}\sqrt{1+\sin\frac{x}{2}}\,dx=\int_0^{2x}\left|\sin\frac{x}{4}+\cos\frac{x}{4}\right|dx$$

$$=4\left[-\cos\frac{x}{2}+\sin\frac{x}{4}\right]_0^{2x}=8$$

Sol.51. **(d)**

$$\int_0^{x/4}\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx=\int_0^{x/4}\left(\frac{\sin x+\cos x}{\sqrt{\sin x.\cos x}}\right)dx$$

$$=\sqrt{2}\int_0^{x/4}\left(\frac{\sin x+\cos x}{\sqrt{1+\sin^2x+\cos^2x-2\sin x.\cos x}}\right)dx$$

$$=\sqrt{2}\int_0^{\pi/4}\left(\frac{\sin x+\cos x}{\sqrt{1-(\sin x-\cos x)^2}}\right)dx$$

put sinx−cosx = t

then (sinx + cosx)dx = dt

$$=\sqrt{2}\int_{-1}^{0}\left(\frac{dt}{\sqrt{1-t^2}}\right)=\sqrt{2}[\sin^{-1}x]_{-1}^{0}=\frac{\pi}{\sqrt{2}}$$

Sol.52. **(a)**

$$\int_0^1 x(1-x)^9dx=\int_0^1 x^9(1-x)dx=\int_0^1(x^9-x^{10})dx$$

$$=\left[\frac{x^{10}}{10}-\frac{x^{11}}{11}\right]_0^4=\left[\frac{1}{10}-\frac{1}{11}\right]=\frac{1}{110}$$

Sol.53. **(a)**

$$\int_a^b x^3dx=\left[\frac{x^4}{4}\right]_a^b=\left[\frac{a^4-b^4}{4}\right]=0$$

$\Rightarrow a^4-b^4=0\Rightarrow a=\pm b$

$$\int_a^b x^2dx=\left[\frac{x^3}{3}\right]_a^b=\left[\frac{a^3-b^3}{3}\right]=\frac{2}{3}$$

$\Rightarrow a^3=b^3+2$

by these equations a = 1 and b = –1

Sol.54. **(c)**

odd function so by direct property answer is zero.

Sol.55. **(a)**

$$\int_0^{\sqrt{2}}[x^2]dx=\int_0^1 0dx+\int_1^{\sqrt{2}}1dx=\sqrt{2}-1$$

Sol.56. **(b)**

$$\int_1^c x\ell n\,x\,dx=\left[\ell nx.\frac{x^2}{4}\right]_1^e-\int_1^e\frac{1}{x}.\frac{x^2}{2}dx$$

$$=\frac{e^2}{2}-\left[\frac{x^2}{4}\right]_1^e=\frac{e^2+1}{4}$$

Sol.57. **(b)**

$\int e^{ax}\sin bx\,dx=\frac{e^{ax}}{a^2+b^2}[a\sin bx-b\cos bx]$

$$\int_0^{\pi}e^x\sin x\,dx=\left[\frac{e^x}{2}(\sin x-\cos x)\right]_0^{\pi}=\frac{e^{\pi}+1}{2}$$

Sol.58. **(d)**

$$\int_{-1}^{1}\left\{\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right)\right\}=\left[\tan^{-1}\frac{1}{x}\right]_{-1}^{1}$$

$=\tan^{-1}1+\tan^{-1}1=\pi/2$

Sol.59. **(d)**

$$\int_2^8|x-5|=-\int_2^5(x-5)+\int_5^8(x-5)$$

$$=-\left[\frac{x^2}{2}-5x\right]_2^5+\left[\frac{x^2}{2}-5x\right]_5^8$$

$$=-\left[\frac{25}{2}-25-2+10\right]+\left[32-40-\frac{25}{2}+25\right]=9$$

Sol.60. **(b)**

$$\int_a^b[x]dx+\int_a^b[-x]dx=\int_a^b kdx+\int_a^b(-1-k)dx$$

$$=\int_a^b(k-1-k)dx=\int_a^b(-1)dx=a-b$$

Sol.61. **(b)**

$$\int_0^{\pi/2}|\sin x-\cos x|dx$$

$$=-\int_0^{\pi/4}(\sin x-\cos x)dx+\int_{\pi/4}^{\pi/2}(\sin x-\cos x)dx$$

$$=-[-\cos x-\sin x]_0^{\pi/4}+[-\cos x-\sin x]_{\pi/4}^{\pi/2}$$

$$=-\left[-\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2}}+1\right]+\left[-1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right]$$

=2(√2-1)

Sol.62. **(b)**

$$\int_0^{\pi/2}e^{\sin x}\cos x\,dx$$

$$=\left[e^{\sin x}\right]_0^{\pi/2}=e^{\sin\pi/2}e^0=e-1$$

Sol.63. **(b)**

$$I_1=\int_0^{\pi}\frac{xdx}{1+\sin x}\text{.............(i)}$$

$$\left[\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$$

$$I_1=\int_0^{\pi}\frac{(\pi-x)dx}{1+\sin(\pi-x)}=\int_0^{\pi}\frac{(\pi-x)dx}{1+\sin x}\text{............(ii)}$$

(i) + (ii)

$$2I_1=\int_0^{\pi}\frac{\pi dx}{1+\sin x}=\pi\int_0^{\pi}\frac{dx}{1+\sin x}$$

$$\left[\int_0^{2a}f(x)dx=2\int_0^{a}f(x)dx \text{ when } f(2a-x)=f(x)\right]$$

$$2I_1=2\pi\int_0^{\pi/2}\frac{dx}{1+\sin x}$$

$$2I_1=2\pi\int_0^{\pi/2}\frac{dx}{1+\cos\left(\frac{\pi}{2}-x\right)}=2\pi\int_0^{\pi/2}\frac{dx}{2\cos^2\left(\frac{\pi}{4}-\frac{x}{2}\right)}$$

$$I_1=\pi\int_0^{\pi/2}\frac{\sec^2\left(\frac{\pi}{4}-\frac{x}{2}\right)dx}{2}=\frac{\pi}{2}\int_0^{\pi/2}\sec^2\left(\frac{\pi}{4}-\frac{x}{2}\right)dx$$

$$I_1=\frac{\pi}{2}\left[\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right]_0^{\pi/2}\times(-2)$$

$$I_1=\frac{\pi}{2}[0-1]\times(-2)=\pi$$

Sol.64. (d)

$$I_2=\int_0^{\pi}\frac{(\pi-x)dx}{1-\sin(\pi+x)}$$

$$\left[\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$$

$$I_2=\int_0^{\pi}\frac{xdx}{1+\sin x}=I_1$$

by previous question

$I_1=\pi$

$I_1+I_2=2\pi$

Sol.65. (b)

$$\int_0^{\pi/4}\left(\tan^3x+\tan x\right)dx$$

$$=\int_0^{\pi/4}\tan x\left(\tan^2x+1\right)dx$$

$$=\int_0^{\pi/4}\tan x\left(\sec^2x\right)dx$$

put tanx = t

$$=\int_0^1 tdt=\left[\frac{t^2}{2}\right]_0^1=\frac{1}{2}$$

Sol.66. (c)

$$I=\int_0^a\frac{f(a-x)}{f(x)+f(a-x)}dx$$

$$\left[\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$$

$$I=\int_0^a\frac{f(x)}{f(x)+f(a-x)}dx$$

2I = 0

I = 0

Sol.67. (a)

$$\int_0^a[f(x)+f(-x)]dx=\int_{-a}^a g(x)dx$$

$$\int_0^a f(x)dx+\int_0^a f(-x)dx=\int_{-a}^0 g(x)dx+\int_0^a g(x)dx$$

Solving $\int_{-a}^0 g(x)dx$ individually

$$\int_{-a}^0 g(x)dx$$

Put x = - t , dx = - dt

$$\int_{-a}^0 g(x)dx=-\int_a^0 g(-t)dx=\int_0^a g(-x)dx$$

$$\int_0^a f(x)dx+\int_0^a f(-x)dx=\int_0^a g(-x)dx+\int_0^a g(x)dx$$

then g(x) = f(x)

Sol.68. (d)

$$\int_2^{10}\frac{f('x)}{f(x)}dx$$

$=[\log f(x)]_2^{10}$

$=\log f(10)-\log f(2)$

$=\log\frac{f(10)}{f(2)}$

$=\log\frac{2^{10}}{2^2}=\log 2^8$

=8log2

Sol.69. (d)

$\int_{-2}^0 f(x)dx=k\ (k\in R)$

Case 1: If graph of f(x) lie above the x axis then |f(x(|= f(x)

$\int_{-2}^0|f(x)|dx=k\ (k\in R)$

Case 2: If graph of f(x) lie below the x axis |f(x)|= –f(x)

$\int_{-2}^0|f(x)|dx=k\ (k\in R^+)$

Case 3: If graph of f(x)
Cuts the x axis
Then Area of |f(x)|
Will be greater than f(x)

$\int_{-2}^0|f(x)|dx>k$

So answer is equal to or greater than k.

Sol.70. (b)

$$\int_1^4 f'(x)dx=[f(x)]_1^4=f(4)-f(1)=0$$

Sol.71. (c)

$$\int_0^{\pi/2}e^{\log\cos x}dx=\int_0^{\pi/2}\cos xdx=[\sin x]_0^{\pi/2}=1$$

Sol.72. (a)

$$I=\int_0^{\pi}\log\left(\tan\frac{x}{2}\right)dx \quad \ldots(1)$$

f(x) = f(a + b – x)

$$I=\int_0^{\pi[}\log\left(\cot\frac{x}{2}\right)dx` \quad \ldots(2)$$

By adding both equations

$$2I=\int_0^{\pi}\left[\log\left(\tan\frac{x}{2}\right)+\log\left(\cot\frac{x}{2}\right)\right]dx$$

$$=\int_0^{\pi}\log\left(\tan\frac{x}{2}.\cot\frac{x}{2}\right)dx$$

$$=\int_0^{\pi}\log/dx=0$$

Sol.73. (b)

$$\int_0^{\pi/4}\frac{dx}{(\sin x+\cos x)^2}$$

$$=\int_0^{\pi/4}\frac{\sec^2x}{(\tan x+1)^2}dx$$

Put tanx = t

Sec^2x dx = dt and charge limit

$$\Rightarrow\int_0^1\frac{1}{(t+1)^2}dt$$

$$=\left[-\frac{1}{t+1}\right]_0^1$$

$=-\frac{1}{2}+1=\frac{1}{2}$

Sol.74. (b)

$$\int_{-2}^{-1}\frac{x}{|x|}dx$$

From –2 to –1

$$\Rightarrow\frac{x}{|x|}=-1$$

$$\Rightarrow\int_{-2}^{-1}(-1)dx=(-x)_{-2}^{-1}$$

= –2 = –1

Sol.75. (b)

$$I=\int_0^1\ln\left(\frac{1}{x}-1\right)dx=\int_0^1\ln\left(\frac{1-x}{x}\right)dx \quad \ldots\ldots(i)$$

$$\int_a^b f(a+b-x)dx=\int_a^b f(x)dx$$

$$I=\int_0^1\ln\left(\frac{x}{1-x}\right)dx \quad \ldots..(ii)$$

Add both equations

$$2I=\int_0^1\ln\left(\frac{1-x}{x}\right)+\ln\left(\frac{x}{1-x}\right)dx$$

$$2I=\int_0^1\ln\left(\frac{1-x}{x}\right)\left(\frac{x}{1-x}\right)dx$$

$$2I=\int_0^1\log 1.dx=0$$

Sol.76. (d)

$$I=\int_0^{20\pi}\left(\sin^4x+\cos^4x\right)dx=20\int_0^{\pi}\left(\sin^4x+\cos^4x\right)dx$$

$$=40\int_0^{\pi/2}\left(\sin^4x+\cos^4x\right)dx=40k$$

Sol.77. (a)

$$\int_{-\pi/2}^{\pi/2}\left(e^{\cos x}\sin x+e^{\sin x}\cos x\right)dx$$

Here $e^{\cos x}\sin x$ is an odd function so

$$\int_{-\pi/2}^{\pi/2}\left(e^{\cos x}\sin x\right)dx = 0$$

And

$$\int_{-\pi/2}^{\pi/2}\left(e^{\sin x}\cos x\right)dx = \left(e^{\sin x}\right)_{-\pi/2}^{\pi/2} = e - \frac{1}{e} = \frac{e^2-1}{e}$$

Solution for three questions

$f(x) = Pe^x + Qe^{2x} + Re^{3x}$

$f(o)$ $P + Q + R = 6$ …(1)

$f'(x) = Pe^x + 2Qe^{2x} + 3Re^{3x}$

$f'(\log 3) = 3P + 18Q + 81R = 282$

$\Rightarrow P + 6Q + 27R = 94$ …(2)

$$\int_0^{\log 2} f(x)dx = \left[Pe^x + \frac{Qe^{2x}}{2} + \frac{Re^{3x}}{3}\right]_0^{\log 2}$$

$=2P + 2Q + 8/3\,R - P - Q/2 - R/3$

$=P + \frac{3}{2}Q + \frac{7}{3}R = 11$ …(3)

By solving above three equations

P=1, Q=2, R=3

Sol.78. (b)

By above solution Q=2

Sol.79. (c)

By above solution R=3

Sol.80. (d)

By above solution

$f'(o) = P + 2Q + 3R$

$=1+4+9=14$

Sol.81. (d)

$$I_1 = \int_0^{\pi}\frac{x}{1+\cos^2 x}dx \quad …(1)$$

$$\int_a^b f(a+b-x)dx = \int_a^b f(x)dx$$

$$I = \int_0^{\pi}\frac{\pi - x}{1+\cos^2 x}dx \quad …(2)$$

(1)+(2)

$$2I = \pi\int_0^{\pi}\frac{x}{1+\cos^2 x}dx$$

$$2I = 2\pi\int_0^{\pi/2}\frac{1}{1+\sin^2 x}dx$$

$$I, II, \int_0^{\pi/2}\frac{1}{1+\sin^2 x}dx \quad …(3)$$

$$I_2 = \int_0^{\pi}\frac{1}{1+\sin^2 x}dx = 2\int_0^{\pi/2}\frac{1}{1+\sin^2 x}dx \quad …(4)$$

By (3) & (4)

$$\int_0^{\pi/2}\frac{1}{1+\sin^2 x}dx = \frac{I_1}{\pi} = \frac{I_2}{2} \quad …(5)$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{\pi}{2} \Rightarrow \frac{I_1+I_2}{I_1-I_2} = \frac{\pi+2}{\pi-2}$$

By equation (3)

Sol.82. (d)

$$I_1 = \pi\int_0^{\pi/2}\frac{1}{1+\sin^2 x}dx = \pi\int_0^{\pi/2}\frac{\sec^2 x\,dx}{1+2\tan^2 x}$$

But $\tan x = t$

$\sec^2 x\,dx = dt$

$$I_1 = \frac{\pi}{2}\int_0^{\infty}\frac{dt}{\left(\frac{1}{\sqrt{2}}\right)^2 + t^2}$$

$$.\ \frac{\pi}{2}\sqrt{2}\left[\tan^{-1}\left(\sqrt{2}t\right)\right]_0^{\infty} = \frac{\pi^2}{2\sqrt{2}}$$

$$I_1 = \frac{\pi^2}{2\sqrt{2}}$$

$$I_1^2 = \frac{\pi^4}{8}$$

$$8\,I_1^2 = \pi^4$$

Sol.83. (a)

By equation (5)

$I_2 = 2/\pi\, I_1$

$$= \frac{2}{\pi}.\frac{\pi^2}{2\sqrt{2}}$$

$I_2 = \pi/\sqrt{2}$

Sol.84. (a)

$$\int_a^b \frac{|x|}{x}dx$$

for $a < 0 < b$

$$\int_a^0(-1)dx + \int_0^b 1dx$$

$$= (-x)_a^0 + (x)_0^b$$

$=a+b$

Sol.85. (b)

$$\int_a^b \frac{|x|}{x}dx$$

for $a < 0 < b$

$$\int_a^b(-1)dx = -(x)_a^b$$

$= -$ [b-a]

$= a - b$

Sol.86. (b)

$$I = \int_{-2\pi}^{2\pi}\frac{\sin^4 x + \cos^4 x}{1+3^x}dx ....(i)$$

$$\int_a^b f(a+b-x)dx = \int_a^b f(x)dx$$

$$I = \int_{-2\pi}^{2\pi}\frac{\sin^4 x + \cos^4 x}{1+3^{-x}}dx ....(ii)$$

add (i) and (ii)

$$2I = \int_{-2\pi}^{2\pi}\left(1+3^x\right)\frac{\sin^4 x + \cos^4 x}{1+3^x}dx$$

$$2I = \int_{-2\pi}^{2\pi}\sin^4 x + \cos^4 x\,dx$$

$$2I = 2\int_0^{2\pi}\sin^4 x + \cos^4 x\,dx$$

$$I = \int_0^{2\pi}\sin^4 x + \cos^4 x\,dx$$

$$I = 4\int_0^{\pi/2}\left(\sin^4 x + \cos^4 x\right)dx$$

$$I = 4\int_0^{\pi/2}\left(1 - 2\sin^2 x\cos^2 x\right)$$

$$I = 4\int_0^{\pi/2}\left(1 - \frac{\sin^2 2x}{2}\right) = 4\int_0^{\pi/2}\left(1 - \left(\frac{1-\cos 4x}{4}\right)\right)$$

$$I = 4\int_0^{\pi/2}\left(\frac{3}{4} + \frac{\cos 4x}{5}\right) = 4\left(\frac{3x}{4} + \frac{\sin 4x}{16}\right)_0^{\pi/2}$$

$$I = 4\int_0^{\pi/2}\left(\frac{3}{4} + \frac{\cos 4x}{5}\right) = 4\left(\frac{3\pi}{8}\right) = \frac{3\pi}{2}$$

$$I = \int_0^{2\pi}\sin^4 x + \cos^4 x\,dx = \frac{3\pi}{2}$$

$$I = 2\int_0^{\pi}\sin^4 x + \cos^4 x\,dx = \frac{3\pi}{2}$$

$$\int_0^{\pi}\sin^4 x + \cos^4 x\,dx = \frac{3\pi}{4}$$

Sol.87. (c)

By above solution

$$I = \int_0^{2\pi}\sin^4 x + \cos^4 x\,dx = \frac{3\pi}{2}$$

Sol.88. (d)

$$\int_0^{8\pi}|\sin x|\,dx = 8\int_0^{\pi}|\sin x|\,dx$$

$$= 8\int_0^{\pi}\sin x\,dx = 8\times 2 = 16$$

Sol.89. (a)

$$\int_{-1}^{0}|x-1|[x]dx = \int_{-1}^{0}\{-(x-1)(-1)\}dx$$

$$= \int_{-1}^{0}(x-1)dx = \left[\frac{x^2}{2} - x\right]_{-1}^{0} = -\frac{3}{2}$$

Sol.90. (d)

$$\int_0^2|x-1|[x]dx = \int_0^1|x-1|[x]dx + \int_1^2|x-1|[x]dx$$

$$= \int_0^1\{-(x-1)\times 0\}dx + \int_1^2\{(x-1)\times 1\}dx$$

$$= 0 + \left[\frac{x^2}{2} - x\right]_1^2 = \frac{1}{2}$$

Sol.91. (a)

$$3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1 \quad ………..(i)$$

replace x with 1/x

$$3f\left(\frac{1}{x}\right) + f(x) = x + 1 \quad …………(ii)$$

by solving above equations

$$f(x) = \frac{3}{8x} - \frac{x}{8} + \frac{1}{4}$$

$= \left[\frac{3}{8}\log 2 - \frac{4}{16} + \frac{2}{4} - 0 + \frac{1}{16} - \frac{1}{4}\right]$

$= \left[3\log 2 + \frac{1}{2}\right] = 3\log 2 + \frac{1}{2}\log e$

$= \log 8 + \log\sqrt{e} = \log(8\sqrt{e})$

Sol.92. (b)

$$I = \int_{-1}^{1} (3\sin x - \sin 3x)\cos^2 x\,dx$$

$$I = \int_{-1}^{1} (4\sin^3 x)\cos^2 x\,dx = 0$$

because above function is odd function

Sol.93. (d)

y = f(x) = e^{-x} is positive for all real value of x

so |f(x)| = f(x)

than p = q

Sol.94. (a)

$$I = \int_0^{\frac{\pi}{2}} \frac{a + \sin x}{2a + \sin x + \cos x}\,dx \quad \text{……..(i)}$$

apply $\int_a^b f(a+b-x)dx = \int_a^b f(x)dx$

$$I = \int_0^{\frac{\pi}{2}} \frac{a + \cos x}{2a + \sin x + \cos x}\,dx \quad \text{……...(ii)}$$

add (i) and (ii)

$$2I = \int_0^{\frac{\pi}{2}} 1\,dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Sol.95. (d)

$$\phi(a) = \int_a^{a+100\pi} |\sin x|\,dx$$

$$\phi(a) = \int_0^{100\pi} |\sin x|\,dx$$

given function is periodic function

$$\phi(a) = 100\int_0^{\pi} |\sin x|\,dx$$

$$\phi(a) = 100\int_0^{\pi} \sin x\,dx = 200$$

Sol.96.

by above solution

$\phi(a) = 200$

$\phi'(a) = 0$

Sol.97. (d)

f(x) = $|x^2 - x - 2|$

f(x) = $|(x-2)(x+1)|$

f(x) is positive for x < –1 and x > 2

f(x) is negative for – 1 < x < 2

$$\int_0^2 f(x)dx = -\int_0^2 (x^2 - x - 2)dx$$

$$= -\left[\frac{x^3}{3} - \frac{x^2}{2} - 2x\right]_0^2$$

$$= -\left[\frac{8}{3} - \frac{4}{2} - 4\right] = \frac{10}{3}$$

Sol.98. (b)

$$\int_1^3 f(x)dx = \int_1^2 f(x)dx + \int_2^3 f(x)dx$$

$$= -\int_1^2 (x^2 - x - 2)dx + \int_2^3 (x^2 - x - 2)dx$$

$$= -\left[\frac{x^3}{3} - \frac{x^2}{2} - 2x\right]_1^2 + \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x\right]_2^3$$

$$= -\left[\frac{8}{3} - \frac{4}{2} - 4 - \frac{1}{3} + \frac{1}{2} + 2\right] + \left[\frac{27}{3} - \frac{9}{2} - 6 - \frac{8}{3} + \frac{4}{2} + 4\right]$$

= 3

Sol.99. (b)

$f(t) = \ln\left(t + \sqrt{1+t^2}\right)$

$f(-t) = \ln\left(-t + \sqrt{1+t^2}\right)$

$f(t) + f(-t) = \ln\left(t + \sqrt{1+t^2}\right) + \ln\left(-t + \sqrt{1+t^2}\right)$

$f(t) + f(-t) = 0$

$f(-t) = -f(t)$ ……..(i)

$g(t) = \tan(f(t))$

$g(-t) = \tan(f(-t)) = \tan(-f(t))$

$= -\tan(f(t)) = -g(t)$

so g(t) is and odd function

and for odd functions

$$\int_{-a}^{a} g(t)dt = 0$$

Sol.100. (c)

$$\int_0^{\pi/2} \frac{1}{g(x)}dx$$

$$= \int_0^{\pi/2} \frac{dx}{\sin x + \cos x + 1}$$

$$= \int_0^{\pi/2} \frac{dx}{\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} + \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}} + 1}$$

$$= \int_0^{\pi/2} \frac{\left(1+\tan^2\frac{x}{2}\right)dx}{2\tan\frac{x}{2} + 1 - \tan^2\frac{x}{2} + 1 + \tan^2\frac{x}{2}}$$

$$= \int_0^{\pi/2} \frac{\left(\sec^2\frac{x}{2}\right)dx}{2 + 2\tan\frac{x}{2}}$$

let $2 + 2\tan\frac{x}{2} = t$

$$= \ln\left[2 + 2\tan\frac{x}{2}\right]_0^{\pi/2} = \ln(2 + 2 - 2) = \ln 2$$

Sol.101. (c)

$$I = \int_0^{\pi/2} \frac{f(x)}{g(x)}dx$$

$$I = \int_0^{\pi/2} \frac{\sin x\,dx}{\sin x + \cos x + 1} \quad \text{……..(i)}$$

apply $\int_a^b f(a+b-x)dx = \int_a^b f(x)dx$

$$I = \int_0^{\pi/2} \frac{\cos x\,dx}{\sin x + \cos x + 1} \quad \text{………(ii)}$$

add both equations

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x\,dx}{\sin x + \cos x + 1}$$

add and subtract 1

$$2I = \int_0^{\pi/2} \frac{(\sin x + \cos x + 1 - 1)dx}{\sin x + \cos x + 1}$$

$$2I = \int_0^{\pi/2} 1 - \frac{1}{\sin x + \cos x + 1}dx$$

$$2I = \int_0^{\pi/2} 1\,dx - \int_0^{\pi/2} \frac{1}{\sin x + \cos x + 1}dx$$

$$2I = \frac{\pi}{2} - \ln 2$$

$$I = \frac{\pi}{4} - \frac{\ln 2}{2}$$

Sol.102. (b)

$$\int_{\sqrt{2}}^{\sqrt{3}} f(x)dx = \int_{\sqrt{2}}^{\sqrt{3}} [x^2]dx = \int_{\sqrt{2}}^{\sqrt{3}} 2dx$$

$$[2x]_{\sqrt{2}}^{\sqrt{3}} = 2\sqrt{3} - 2\sqrt{2}$$

Sol.103. (a)

$$\int_{\sqrt{2}}^{2} f(x)dx = \int_{\sqrt{2}}^{\sqrt{3}} f(x)dx + \int_{\sqrt{3}}^{2} f(x)dx$$

$$\Rightarrow \int_{\sqrt{2}}^{\sqrt{3}} [x^2]dx + \int_{\sqrt{3}}^{2} [x^2]dx \Rightarrow \int_{\sqrt{2}}^{\sqrt{3}} 2dx + \int_{\sqrt{3}}^{2} 3dx$$

$$\Rightarrow [2x]_{\sqrt{2}}^{\sqrt{3}} + [3x]_{\sqrt{3}}^{2}$$

$$= 2\sqrt{3} - 2\sqrt{2} + 6 - 3\sqrt{3} = 6 - \sqrt{3} - 2\sqrt{2}$$

Sol.104. (c)

$f(x) = x - [x]$

This function is called fractional part of x whose time period is 1.

$$\int_n^{n+1} (x - [x])dx = \int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

Sol.105. (a)

Let

ln x = t

x = e^t

dx = e^t dt

$$I_1 = \int_e^{e^2} \frac{dx}{\ln x} = \int_1^2 \frac{e^t}{t}dt = \int_1^2 \frac{e^x}{x}dx = I_2$$

So $I_1 - I_2 = 0$